



Research paper

Investigating the impact of numerical uncertainties in CFD-generated training data on neural network models for wind-turbine wake predictions

Maokun Ye^a, Yao Zhong^b, Hamn-Ching Chen^c, Decheng Wan^{a,*}^a Computational Marine Hydrodynamics Lab (CMHL), School of Ocean and Civil Engineering, Shanghai Jiao Tong University, Shanghai, China^b Zhongnan Engineering Corporation Limited, Changsha, China^c Zachry Department of Civil & Environmental Engineering, Texas A&M University, College Station, TX, USA

ARTICLE INFO

Keywords:

Wind turbine wake
CFD
Neural networks
Numerical uncertainties
PINN

ABSTRACT

Neural networks (NNs) are increasingly applied to model complex flow fields, including wind turbine wakes. Current studies often employ full-field CFD-generated data for NN training, but two critical concerns remain unaddressed: (1) real-world measurements are limited to sparse sensor arrays (e.g., anemometers or lidar lines), and (2) the impact of numerical uncertainties of CFD-generated data on NN models' performance remains underexplored. In particular, the limited measurements may amplify the impact of data-embedded numerical uncertainties. To address this issue, in this work, we investigate how numerical uncertainties in CFD-generated training data affect the performance of NN models for wind turbine wake predictions. We first generated wind turbine wake datasets with varying spatial and temporal resolutions using CFD simulations and conducted a thorough numerical uncertainty analysis. A "weakly-constrained PINN" was then developed to predict full wind turbine wake fields from sparse training data. In total 9 NN models were trained using CFD-generated datasets with different grid-time configurations, and their performance was evaluated on wake profile (sparse line) and full-field prediction tasks. Results show that training data resolution significantly impacts NN performance, with spatial accuracy being more critical than temporal accuracy for the certain case investigated in the current work. CFD data with lower numerical uncertainties yield NN models with higher prediction accuracy ($R^2 > 0.9$), while CFD data with higher numerical uncertainties lead to poor performance ($R^2 < 0.8$). Notably, downstream distance influences prediction accuracy, with far-field wake predictions outperforming near-field results. The study emphasizes the need to quantify numerical uncertainties in data-driven wake modeling, and provides practical guidance on balancing CFD resolution and computational cost for generating training datasets for NN models.

1. Introduction

Wind energy has emerged as a pivotal renewable energy source in the global transition toward sustainable power systems. The efficiency of wind farms, however, hinges critically on the accurate prediction of wind turbine wake effects, which directly influence power-generation forecasts and, consequently, the economic viability of wind energy projects (Archer et al., 2018). Wake models, which describe the spatial distribution of the flow quantities in the wake field, are indispensable tools for optimizing turbine placement and operational strategies. Despite their importance, the accuracy and computational efficiency of these models remain significant challenges (Porté-Agel et al., 2020). Developing precise yet computationally tractable wake calculation methods is therefore essential to maximize energy output and minimize

financial risks in wind farm development.

Existing approaches for wind-turbine wake modeling can be broadly categorized into analytical wake models and computational fluid dynamics (CFD) simulations. Analytical wake models, such as the Jensen (1983) or Bastankhah models (Bastankhah and Porté-Agel, 2014), rely on simplified empirical or semi-empirical formulations to approximate wake characteristics. While these models are computationally efficient and widely adopted in industry, their reliance on idealized assumptions often leads to inaccuracies in complex atmospheric conditions or heterogeneous turbine layouts (Amiri et al., 2024). These limitations underscore the trade-off between efficiency and accuracy inherent in traditional wake modeling frameworks. In contrast, CFD simulations resolve the Navier-Stokes equations to capture the intricate physics of turbine wakes with high fidelity. These simulations have been

* Corresponding author.

E-mail address: dcwan@sjtu.edu.cn (D. Wan).<https://doi.org/10.1016/j.oceaneng.2025.123200>

Received 1 July 2025; Received in revised form 10 October 2025; Accepted 14 October 2025

Available online 24 October 2025

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successfully applied to study wake interactions (Deskos et al., 2020), (Ye et al., 2024a), atmospheric boundary layers (Churchfield et al., 2012), (Abkar and Porté-Agel, 2014), and terrain effects (Kuo et al., 2016), providing detailed insights into flow fields. Nevertheless, the computational cost of CFD simulations is prohibitive for large-scale or long-term wind farm analyses, rendering it impractical for real-time operational decisions or iterative design optimizations. This gap between accuracy and computational feasibility has driven the exploration of alternative methodologies.

Recent advancements in artificial neural networks (NNs) have offered new possibilities for wake modeling tasks. Researchers have proposed NN-based frameworks to predict wake characteristics using input parameters such as inflow conditions, turbine geometry, and atmospheric stability. Representative studies include those by Ti et al. (2020), (Ti et al., 2021), who utilized actuator disk model-based CFD simulations to model fixed single-turbine wake fields and established a mapping relationship between turbine inflow conditions, spatio-temporal wake coordinates (x, y, z, t) and flow quantities, i.e. (u, v, w), using a multilayer perceptron (MLP). Zhang et al. (2021) leveraged the superior capability of convolutional neural networks (CNNs) in capturing spatio-temporal flow features, constructing a mapping relation between historical and future wake fields of fixed single turbines. Their network architecture takes sequential time-step wake fields as input and predicts subsequent wake fields. Li et al. (2023) implemented a graph neural network (GNN) with global spatial aggregation capabilities for parameterizing fixed single-turbine wakes, where inputs included inflow velocity and yaw angle, while outputs represented time-averaged steady-state wake fields. The trained model was then integrated with wake superposition models to investigate wake interactions in turbine arrays. Additional similar studies will not be detailed here, and the readers are encouraged to refer to one of our review paper for more information (Ye et al., 2025).

However, all the aforementioned studies leveraged full-field CFD data for training, assuming spatially resolved datasets that are rarely available in real-world scenarios. In practice, field measurements are sparse and restricted to a few anemometers or lidar lines, making it infeasible to generate the dense, spatially resolved datasets required for robust NN training. This discrepancy has motivated efforts to develop physics-informed neural networks (PINNs) (Raissi and Karniadakis, 2018), (Raissi et al., 2019) that integrate physical constraints to enhance generalization from limited data. In wind modeling research, Zhang and Zhao's foundational work (Zhang and Zhao, 2021) proposed the integration of LiDAR-derived datasets with PINNs. Their methodology involved training the model with sparsely distributed horizontal wind field measurements, incorporating physical constraints through 2D Navier-Stokes equations. Validation against large eddy simulation (LES) datasets revealed good agreement between neural network predictions and numerical results. Subsequent investigations expanded this paradigm: Wang et al. (2024) specialized in yaw-induced wake trajectory analysis through analogous PINN architectures, while Zhang and Zhao's later innovation (Zhang and Zhao, 2023) further enhanced the framework's fidelity by implementing full 3D Navier-Stokes formulations coupled with an actuator disk model (ADM). This advancement introduced momentum source terms to mathematically represent turbine blade aerodynamics within the physical loss function, enabling accurate multi-turbine wake predictions at hub-height elevation through optimized neural network training.

Two key insights emerge from existing literature: First, CFD simulations provide a controlled environment for generating high-quality training data across diverse inflow parameters, making them indispensable for developing data-driven wake models. Second, physics-informed approaches like PINNs show significant promise for reconstructing full wake fields from sparse measurements by leveraging physical laws as regularization. However, a critical gap remains: the impact of numerical uncertainties inherent in CFD-generated training data on NN model performance remains underexplored. This oversight is

particularly consequential in sparse-data cases (i.e. real-world tasks), where limit measurements may significantly amplify the influence of data quality on the trained NN models. More specifically, numerical errors in CFD-predicted velocity profiles due to coarse spatial resolution could propagate into trained NN models, harming their predictive capability in real-world deployments.

The aim of this work is to investigate how numerical uncertainties in CFD-generated training data affect the accuracy of NN models for wind-turbine-wake prediction. We focus specifically on scenarios where models are trained using sparse line measurements - a realistic but challenging condition where data quality limitations are most pronounced. With systematical quantification of discretization errors across 9 CFD datasets with varying spatial and temporal resolutions, we then evaluate their effects on the final performance of trained NNs in both localized wake profile and full-field reconstruction tasks. A general diagram to outline and clarify the process of this study is provided in Fig. 1.

The main contributions and novel aspects of this work are summarized as follows.

1. A systematic quantification of numerical uncertainties in CFD-predicted wind turbine characteristics was conducted, addressing both spatial and temporal discretization errors.
2. A weakly-constrained physics-informed neural network (PINN) was developed to accurately reconstruct the full wake field using highly sparse line measurements. Furthermore, the impact of numerical uncertainties inherent in CFD data on the performance of the trained PINN models was thoroughly examined.
3. Our findings underscore the importance of quantifying numerical uncertainties in data-driven wake modeling. These insights offer practical guidance for optimizing the trade-off between CFD resolution demands and computational expenses during the generation of training datasets for data-driven wake models.

The remainder of this paper is organized as follows: Section 2 details the methodologies, including the CFD framework, numerical uncertainty quantification approach, and the weakly-constrained PINN model used in the current work. Section 3 describes the numerical generation of training datasets, while Section 4 presents the verification of CFD uncertainties. Sections 5 and 6 discuss NN training and the performance evaluations of the trained NN models, respectively. Finally, conclusions and recommendations based on the results and discussions are provided in Section 7.

2. Methodologies

2.1. CFD approach

Computational Fluid Dynamics (CFD) simulations are used to generate data for NN training. The governing equations and discretization schemes are described in details in this section.

2.1.1. Governing equations

The unsteady incompressible Reynolds-Averaged Navier-Stokes (RANS) equations are adopted in this work as the governing equations of the flow field around and behind a wind turbine:

$$\frac{\partial U_i}{\partial x_i} = 0 \quad (1)$$

$$\frac{\partial U_i}{\partial t} + U_j \frac{\partial U_i}{\partial x_j} = f_i + \frac{\partial}{\partial x_j} \left(-\frac{p}{\rho} \delta_{ij} + 2\nu S_{ij} - \langle u_i u_j \rangle \right) \quad (2)$$

where i and j equal 1, 2, and 3, representing the 3-dimensional flow field we are solving. U_i is the i -component of the mean flow velocity, and u_i the i -component of the turbulence velocity that needs to be modeled

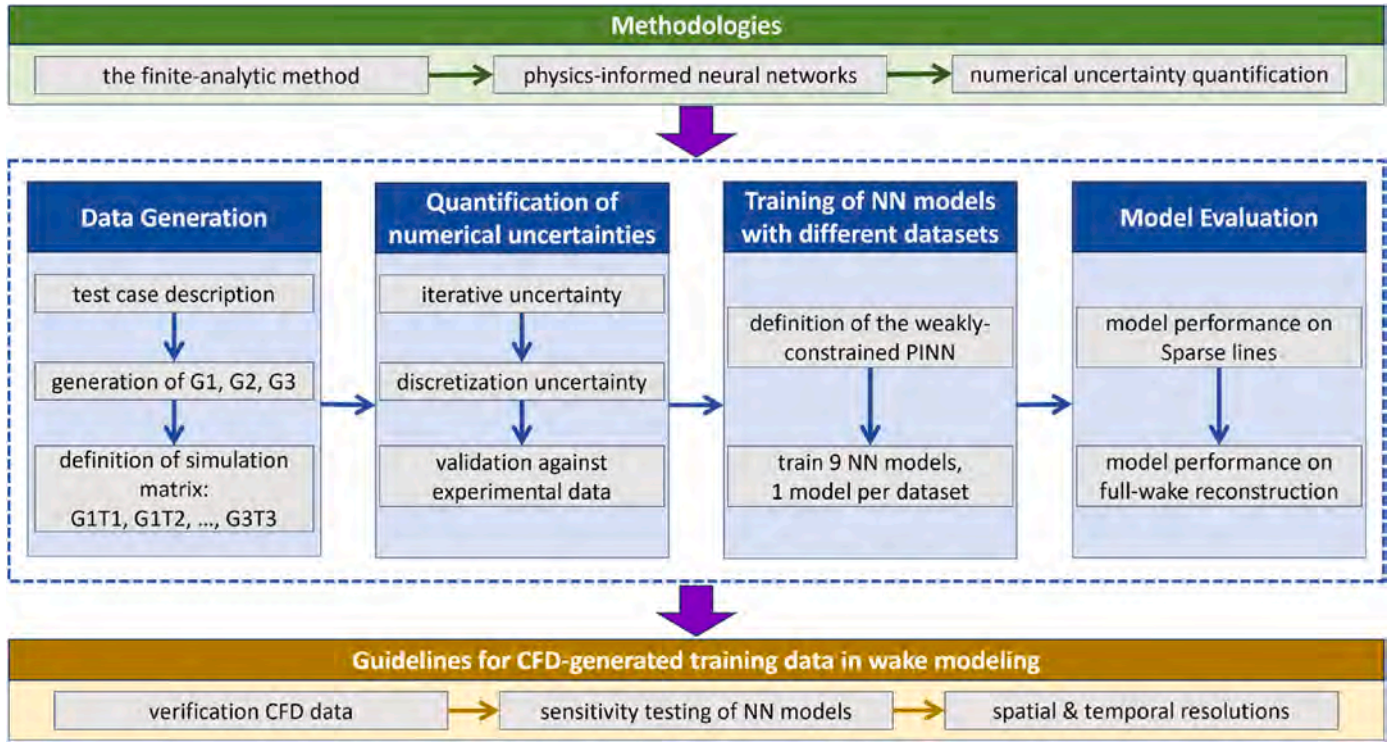


Fig. 1. Outline of the structure of this study.

under the framework of RANS. $\langle u_i u_j \rangle$ is the well-known Reynolds-stress term and it is linked to the mean flow velocity by adopting the eddy viscosity assumption:

$$-\langle u_i u_j \rangle = \nu_t \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) - \frac{2}{3} k \delta_{ij} \quad (3)$$

where ν_t is the so-called eddy viscosity which is an artificial quantity that serves as a representation of the influence of fluid turbulent motion on the mean flow. ν_t is determined by another two transport equations of turbulent kinetic energy k and turbulent dissipation ϵ in the standard $k-\epsilon$ model (Launder and Spalding, 1983). For the two-layer $k-\epsilon$ model (Chen and Patel, 1988) used in this work, the standard $k-\epsilon$ model (Launder and Spalding, 1983) is applied to the flow outside a pre-defined near-wall region, while an explicit formulation for ϵ is used in the region adjacent to solid surfaces:

$$\epsilon = \frac{k^{3/2}}{l_\epsilon} \quad (4)$$

and the turbulent viscosity is:

$$\nu_t = C_\mu \sqrt{k} l_\mu \quad (5)$$

In the above two equations, l_ϵ and l_μ are respectively the spatial scales of ϵ and ν_t , and C_l is defined by $C_l = \kappa C_\mu^{-3/4}$ in which κ is the von Karman constant. More details of this turbulence model can be found in (Chen and Patel, 1988).

2.1.2. The finite-analytic discretization of the governing equations

The incompressible unsteady fluid governing equations are solved by using the finite-analytic (FA) discretization method (Chen, 1982), (Chen and Chen, 1984), (Chen et al., 1990), (Pontaza et al., 2005). The finite-analytic (FA) method is conceptually different from the mainstream discretization methods, i.e. the finite-difference (FD), finite-volume (FV), and finite-element (FE) methods that are usually used in computational physics. In the FA method, the initial computational domain of the problem is decomposed into a number of small

subdomains, and then the analytic solution is derived for the small subdomains by local linearization of the initial PDEs, e.g., the Navier-Stokes equations. The local analytic solution is then leveraged to establish the relations which link the value of one computational point with its adjacent points. Therefore, by leveraging the analytic solutions of the locally linearized PDEs, the initial governing equations can then be readily written in terms of algebraic equations to be solved (Chen, 1982), (Chen et al., 1995).

Eq. (2) can be rewritten as a general transport equation for ϕ , in which ϕ representing velocity component U , V , or W :

$$\phi_{xx} + \phi_{yy} + \phi_{zz} = \text{Re}(\phi_t + U\phi_x + V\phi_y + W\phi_z) + f \quad (6)$$

In the most general form, at any time instance t^n , a center point $P(i, j, k)$ with a spatial coordinate of (i, j, k) of a local element is surrounded by its 26 adjacent points, i.e. $P(i-1, j-1, k-1)$ to $P(i+1, j+1, k+1)$. To linearize the governing PDEs, $U = U_p$, $V = V_p$, and $W = W_p$, can be specified in a small local element. Therefore, the locally linearized transport equation is of the form:

$$\phi_{xx} + \phi_{yy} + \phi_{zz} = \text{Re}(\phi_t + U_p\phi_x + V_p\phi_y + W_p\phi_z) + f_p \quad (7)$$

where U_p , V_p , and W_p are constant values specified at $P(i, j, k)$ to represent the velocity within a small element, and f_p is the source term evaluated at the interior point $P(i, j, k)$.

Then, the time derivative of the variable ϕ , i.e. ϕ_t , can be discretized by a temporal discretization scheme, e.g., a backward finite-difference scheme. As to the boundary conditions for the 27-point element (consists of one interior point $P(i, j, k)$ and its 26 surrounding points), it has 9 nodes on each of the 6 surfaces. The 3D exponential and linear boundary conditions extended based on 2D cases (Chen, 1982), (Chen et al., 1990) can be implemented to approximate the boundary conditions on all the 6 boundary surfaces. The linearized PDEs are then solved analytically within the small element by the method of separation of variables. Here, suppose that the center point of a local element is denoted as $P(i, j, k)$, the final formula describing the relation between $P(i, j, k)$ and its adjacent points is as follows for a 3D time-dependent problem if a second order

temporal scheme is implemented:

$$\phi_p = \frac{1}{1 + (Re/\tau)C_p} \left[\sum_{nb=1}^{26} C_{nb}\phi_{nb} + \frac{Re}{\tau}C_p(1.5\phi_p^{n-1} - 0.5\phi_p^n) - C_p f_p \right] \quad (8)$$

Note that in the above equation, the superscripts of quantities at t^n is omitted and only the quantities at t^{n-1} are marked with the temporal coordinate. 'nb' represents the neighboring computational points adjacent to the center point of $P(i,j,k)$. C_p and C_{nb} are the FA coefficients, and the detailed expressions of them are given by (Chen et al., 1990). It should be mentioned here that in addition to the 27-point discretization, a 19-point, 11-point, or a 7-point, discretization scheme of the Navier-Stokes equation can also be used in the FA method. Detailed introductions of those schemes can be found in (Chen et al., 1990).

In this study, the FANS (Finite-Analytic Navier-Stokes) code is adopted as the numerical solver to solve the fluid governing partial differential equations, i.e. the RANS equations in this study. Due to the fact that the finite-analytic method can only be applied to structured grids, therefore, FANS utilizes the overset grid, or chimera grid, to handle complex geometries. The code solves the governing partial differential equations under a general curvilinear framework within individual overset grid blocks by using body-fitted coordinates. The flow information obtained from different grid blocks is exchanged within the overlapping regions by variable interpolation. A hybrid pressure-velocity coupling algorithm that combines the PISO and the SIMPLER algorithms are developed (Pontaza et al., 2005) and implemented in the code. The FANS solver has been extensively validated and effectively implemented in certain research applications (Huang et al., 2011), (Huang and Chen, 2020), (Huang et al., 2022), (Chen and Chen, 2023). Further, in previous studies, the authors have successfully applied the FANS code in the calculations of wind turbine performance (Kun Ye et al., 2023), (Ye et al., 2024b). Therefore, in the current work, we extend and highlight the capability of the adopted numerical framework in the simulations of wake characteristics of wind turbines.

2.2. Discretization uncertainties in numerical simulations

To confirm the results obtained by numerical simulations, typically two different levels of concerns need to be addressed. The first level is to prove that the governing equations themselves are solved in a proper way, this process is referred to as verification. Then, under this premise, the numerical predictions can be legitimately compared to the experimental measurements to show that the governing equations we derived aiming at this particular problem is correct, i.e. describing the phenomenon properly.

In numerical investigations, three primary classifications of errors are frequently experienced, namely the round-off, iterative, and discretization errors (Coleman and Stern, 1998), (Stern et al., 2001), (Stern et al., 2006), (Eça et al., 2019). The total error ϵ_n encompassing all the three components can therefore be written as:

$$\epsilon_n = \epsilon_{ro} + \epsilon_{it} + \epsilon_d \quad (9)$$

in which.

- ϵ_{ro} is the round-off error arising from the limited floating-point accuracy of machines and can be treated negligible;
- ϵ_{it} is the iterative error which stems from the non-linearity of PDEs and can be justified small if strict convergence criterion is reached during the calculation;
- ϵ_d is the discretization error, and it originates from the reformulating of the governing equations to their algebraic forms. It is the main source contributing to the total numerical error.

As mentioned earlier, the premise of showing a numerical prediction to be correct is to prove the governing equations are solved properly. Typically, this goal is achieved by a systematical procedure of quanti-

fying the discretization error ϵ_d , i.e. the verification procedure. In this work, we follow the framework of conducting the verification study that was proposed by (Eça and Hoekstra, 2014) in which ϵ_d is formulated in a power series:

$$\epsilon_d \simeq \delta_d = \phi_{ij} - \phi_0 = \alpha_x h_i^{p_x} + \alpha_t t_j^{p_t} \quad (10)$$

in which.

- ϕ_{ij} is any quantity derived numerically from grid i and time increment j ;
- ϕ_0 is the "theoretical" solution estimated in the verification procedure through a least-square fitting;
- p_x and p_t are the observed orders of spatial and temporal convergence, respectively;
- h_i and t_j are the relative grid and time step sizes, respectively. Note that in numerical calculations adopting structured grids, $h_i = \frac{\xi_{cell\ 1}}{\xi_{cell\ i}} = \frac{\eta_{cell\ 1}}{\eta_{cell\ i}} = \frac{\zeta_{cell\ 1}}{\zeta_{cell\ i}}$, where $\xi_{cell\ i}$, $\eta_{cell\ i}$ and $\zeta_{cell\ i}$ are the number of grid points in revolving ξ , η , and ζ directions, respectively.

Afterward, the 5 variables in Eq. (10), i.e. ϕ_0 , α_x , p_x , α_t , and p_t , are then calculated by fitting at least 5 data points (preferable at least 1 more point according to Eça and Hoekstra) by the method of least-square, and the data points are obtained by using different combinations of h_i and t_j :

$$S_{RE}(\phi_0, \alpha_x, p_x, \alpha_t, p_t) = \sqrt{\sum_{i=1}^{n_g} \sum_{j=1}^{n_t} w_{ij} \left[\phi_{ij} - \left(\phi_0 + \alpha_x h_i^{p_x} + \alpha_t t_j^{p_t} \right) \right]^2} \quad (11)$$

in which.

- n_g and n_t are the quantity of different computational grids and computational time, respectively;
- w_{ij} represents the weight allocated to a certain CFD result, enabling differentiation among various h_i and t_j pairs based on their comparative resolution, or importance.
- ϕ_0 is the estimated exact solution derived by a least-squares fitting.

This procedure of quantifying the discretization errors differs from the traditional ones (e.g. (Celik et al., 2008)), in that it leverages the least-square method in solving the error equation (Eça and Hoekstra, 2009), and has been shown more robust than the traditional ones in many applications (van der Kolk et al., 2020), (Burmester et al., 2020).

2.3. Physics-informed neural networks

Physics-Informed Neural Networks (PINNs) (Raissi and Karniadakis, 2018) are a hybrid approach that integrates deep learning with physical modeling, particularly effective for solving partial differential equations (PDEs) and other physics-based problems. The core idea of PINNs lies in embedding physical constraints directly into the NN architecture, enabling the model to learn system behavior while adhering to governing physical laws. This method proves especially valuable in scenarios with scarce data or complex problem domains, offering a powerful numerical tool for scientific and engineering applications. A PINN's total loss function combines two components: a data-driven loss term (denoted as $Loss_{label}$) and a physics-based loss term (denoted as $Loss_{pde}$), formulated as:

$$Loss_{total} = Loss_{label} + factor \times Loss_{pde} \quad (12)$$

Here, *factor* is a weighting coefficient that balances the contributions of data fidelity and physical consistency.

While conventional PINNs often incorporate full governing equations (e.g., Navier-Stokes equations) into $Loss_{pde}$, simplified physics can

also be adopted to reduce computational complexity. For instance, Zhou (Zhou et al., 2022) demonstrated that substituting analytical wake models (AWMs) for full PDEs in $Loss_{pde}$ yields robust predictions. Building on this concept, the current study employs a “weakly-constrained PINN” framework for wind turbine wake reconstruction. Specifically, we simplify the physical loss to enforce mass conservation (i.e. divergence-free criterion of incompressible flow) via the continuity equation:

$$Loss_{pde} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \quad (13)$$

where u , v , and w are velocity components. The partial derivatives are computed using automatic differentiation - a standard feature in any deep learning libraries - which enables efficient evaluation of PDE residuals during training. By integrating this physics-based regularization into a multi-layer perceptron (MLP), the network parameters are optimized not only to fit sparse measurement data but also to satisfy continuity constraints across the flow field.

Fig. 2 illustrates the architecture of the weakly-constrained PINN model. The network processes spatiotemporal coordinates (x, y, z, t) as inputs and outputs velocity components (u, v, w) . The hidden layers employ the hyperbolic tangent ($\tanh$) activation function to introduce nonlinearity while maintaining smooth gradient propagation. As shown in Fig. 3, each hidden-layer node computes a weighted sum of inputs followed by activation:

$$Output = \sigma \left(b + \sum_{i=1}^m w_i x_i \right) \quad (14)$$

Where σ denotes the activation function, b is the bias term, and w_i are trainable weights. During training, the backpropagation process iteratively adjusts these weights by minimizing $Loss_{total}$. This optimization process ensures that the learned model captures both the empirical patterns in sparse measurements and the underlying fluid dynamics principles governing wake evolution.

3. Numerical generation of high-fidelity data sets

3.1. Test-case description

Starting from 2008, Norwegian University of Science and Technology (NTNU) conducted successive wind tunnel experiments with

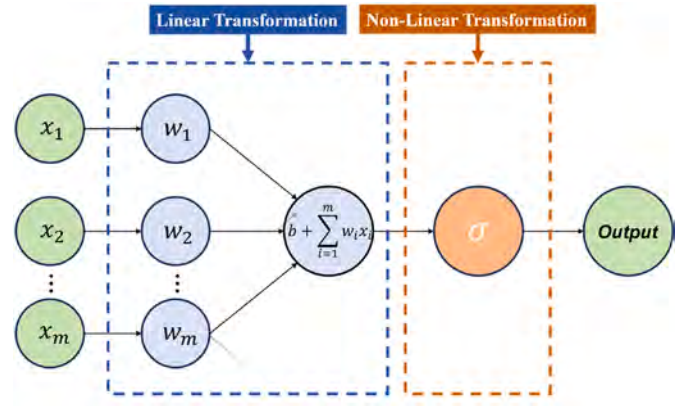


Fig. 3. Illustration of an MLP node.

different setups and initial parameters (Krogstad and Eriksen, 2013), (Krogstad et al., 2015), (Krogstad and Sætran, 2015), (Bartl and Sætran, 2017), (Mühle et al., 2018). In the first wind tunnel test of a single wind turbine, i.e. the NTNU BT1 wind tunnel test, a wind tunnel experimental evaluation was conducted on a three-bladed wind turbine, which was designed with the implementation of the NREL S826 airfoil. The physical sizes of the wind tunnel and the wind turbine in the NTNU BT1 wind tunnel test is listed in Table 1.

Fig. 4 shows the configuration of the NTNU BT1 wind turbine test.

The experimental conditions of the wind tunnel test are listed in Table 2. Note that the Reynolds number (Re) is determined at the blade tip, and the design TSR is adopted in the calculations. The choice of 6 for this particular TSR value is based on two factors: (1) the wind turbine

Table 1

Dimensions of the wind tunnel and the wind turbine in the NTNU BT1 wind tunnel test.

Quantity	Value	Unit
Length of wind tunnel	11.15	m
Width of wind tunnel	2.71	m
Height of wind tunnel at inlet	1.801	m
Height of wind tunnel at outlet	1.851	m
Rotor diameter (D)	0.894	m

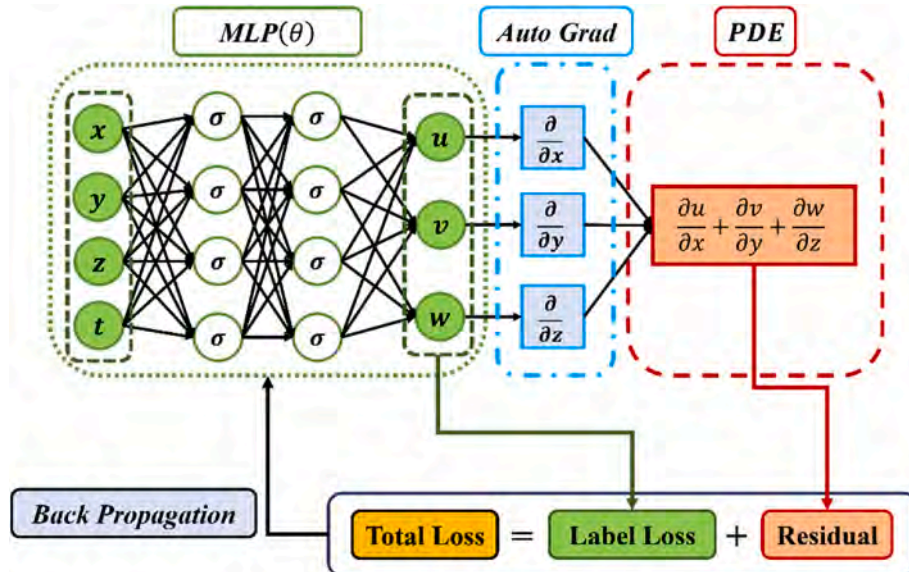


Fig. 2. Illustration of the weakly-constraint PINN.

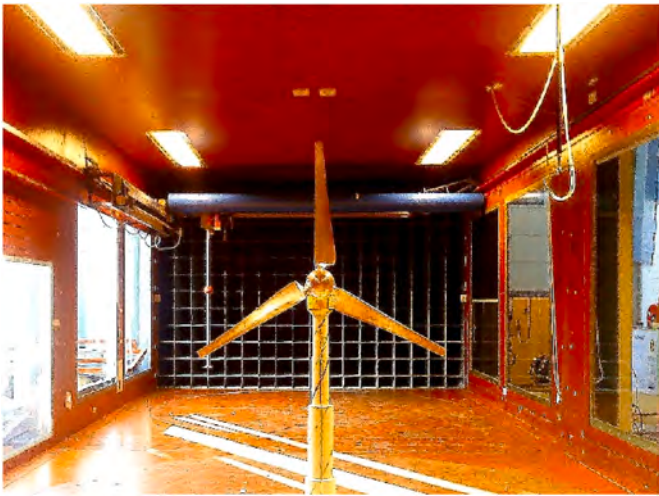


Fig. 4. Experiment setup, adopted from (Krogstad and Eriksen, 2013).

Table 2

Experimental conditions of NTNU BT1 wind tunnel test.

Quantity	Value	Unit
Inflow velocity	10	m/s
Turbulence intensity (TI)	0.3	%
Tested tip-speed ratios ($TSRs$)	1–12	–
Design TSR	6	–
Reynolds number (Re) at the Design TSR	1.036×10^5	–

under investigation has a design TSR of 6, and numerous past studies have conducted simulations at this specific TSR value. As a result, it will be convenient to gather CFD results from alternative numerical methods for comparison. (2) The aim of this study is to demonstrate the influence of the numerical uncertainties in the CFD-generated data on the trained NNs, therefore, a single operating condition of the turbine should be adequate to fulfill this objective.

3.2. CFD domain

A depiction of the computational domain is presented in Fig. 5. The definition of the coordinates is: the origin, i.e. (0,0,0), is located at the centroid of the three blades, the x -axis is normal to the rotating plane of the wind turbine and points towards the outlet, while the y -axis is normal to the ground and points upwards. Furthermore, the figure provides details on the boundary names and the precise dimensions of the computational domain.

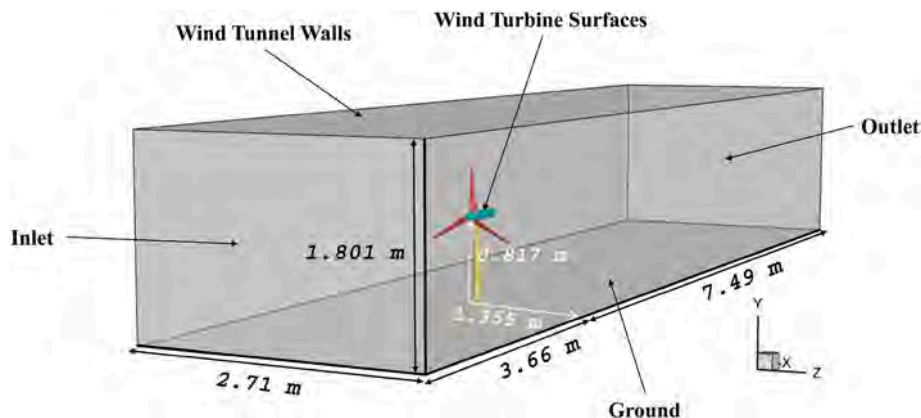


Fig. 5. Illustration of computational domain.

Subsequently, as depicted in Fig. 6, the complete computational domain is decomposed conceptually into three mutually overlapped subdomains, namely, a rotating domain (cyan), a wake domain (yellow), and a background domain (white). Note that the rotor is colored in red. In this manner, we can easily adjust the grid spacings of a selected subdomain without affecting the others. This method is especially suitable for the current study because only the grid resolution of the wake domain is critical for the CFD-predicted wind turbine wake (Ye et al., 2023a), while the impact of grid spacing in the near field is of lower priority (Ye et al., 2021), (Ye et al., 2022a), (Ye et al., 2022b), (Ye et al., 2023b). Therefore, this approach allows us to avoid the need to refine the whole domain simultaneously when only the numerical uncertainty of the CFD-predicted wake is of interest.

3.3. Generation of computational grids

Next, structured overset grids are created within each of the three aforementioned subdomains. The details of the grid generation process are as follows.

3.3.1. Generation of boundary layer grids

First, overset grid blocks are generated to revolve the geometrical details of the wind turbine including the rotor and the stationary parts. In the generation procedure, the first-layer cells are generated in compliance with the criterion of $y^+ < 1$. Fig. 7(a) shows the generated boundary layer grids. In total 39 grid blocks, including three phantom grid blocks used for hole-cutting purposes, are generated for the boundary layers. As a result, 6.03 million grid points are generated for the boundary layer blocks. It should be pointed out here that although not provided in this work, a systematic verification study has already performed in the assessment of errors and uncertainties of the calculated

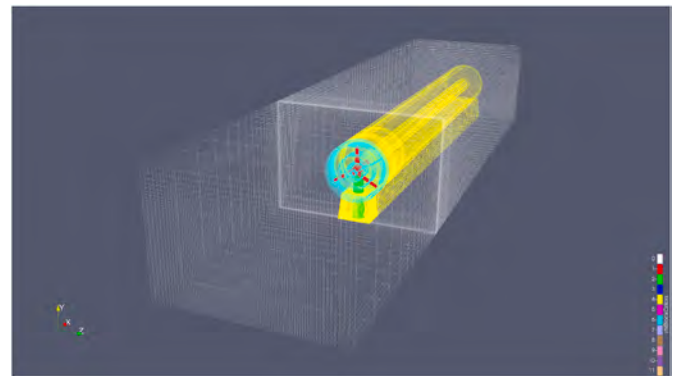


Fig. 6. Illustration of the different components of the computational domain.

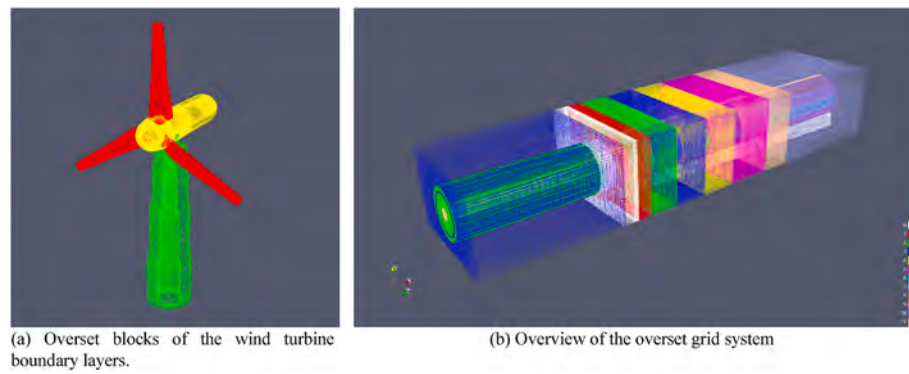


Fig. 7. Overset grid blocks.

thrust and torque of the same turbine, and has been reported by the authors in a previous work (Ye et al., 2024b). Therefore, the grids are generated according to our previous work such that it reaches a balance between the prediction accuracy of the turbine's performance and computational cost.

3.3.2. Generation of wake grids

In this study, we are specifically interested in resolving the wake of the wind turbine, therefore, three sets of numerical grids, i.e. G1, G2, and G3, for the wake subdomain are generated with G1 being the grid with smallest grid spacing and G3 with the largest. Each of the grids is composed of 126 grid blocks in total, with an additional 6 phantom grid blocks specifically employed for cutting holes in the FANS calculations.

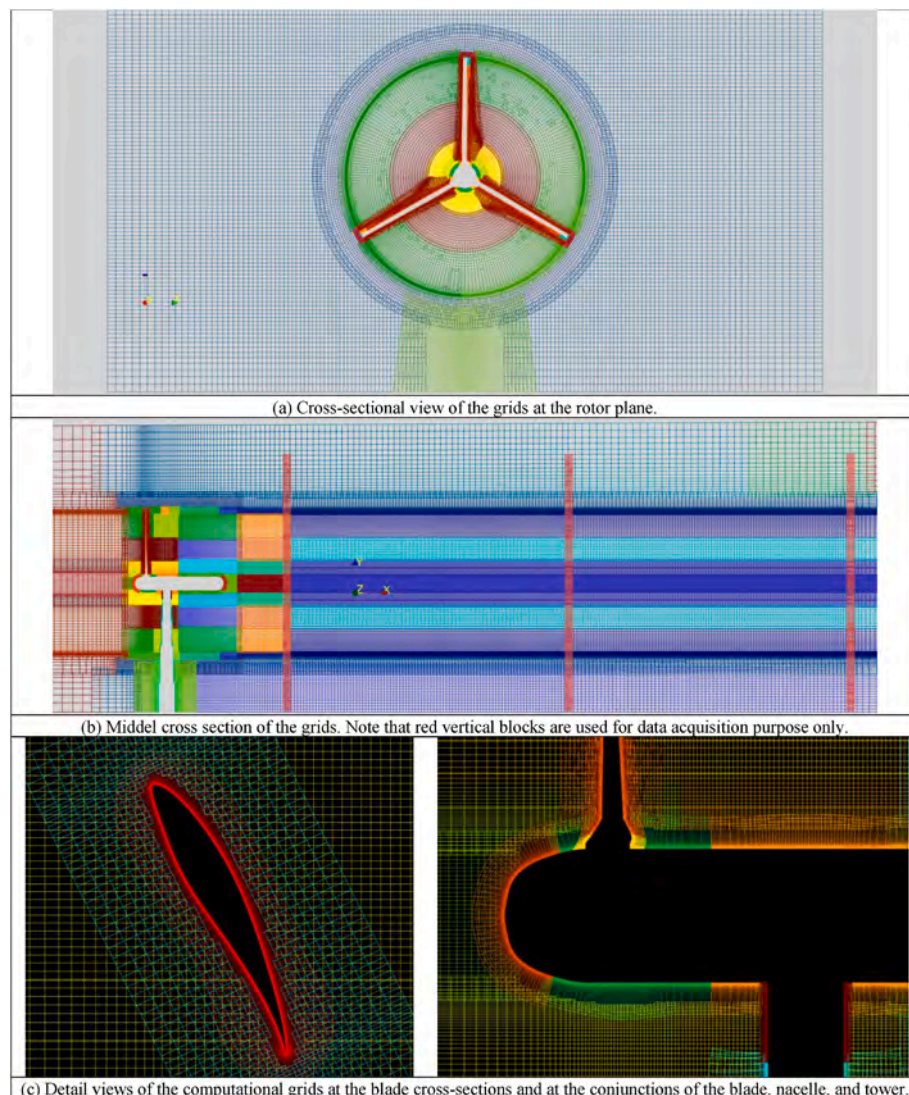


Fig. 8. Details of the computational grids.

Consequently, the total number of computational points of wake subdomains within G1, G2, and G3 is 18.28 million, 10.58 million, and 6.12 million, respectively.

3.3.3. Assembly of overset grid blocks

In this stage, the background grids are generated, and then the three individually generated grids, i.e. the boundary layer grids, wake grids, and the background grids, are assembled to form the complete computational grid. Subsequently, the overall quantity of computational points in the CFD simulations is 24.31, 16.77, and 12.53 million for grid G1, G2, and G3, respectively. Fig. 7(b) shows an overview of the generated computational grid G3. Details of the generated computational grids are shown in Fig. 8: (a) a cross-sectional view at the rotor plane, (b) a middle cross-section, and (c) detailed views of the conjunctions between different overset grid blocks.

Then, those 126 blocks are distributed into 80 cores (2 nodes on the HPC of SJTU) to be calculated in parallel, and the number of grid points in each core should be comparable to ensure that CPU cores finish the computations in approximately the same timeframe in parallel calculations. The boundary condition files of the 126 blocks are respectively defined, with six boundaries for each block. These conditions are also distributed among the 80 cores for parallel computation. Furthermore, the connection relationships of the 126 overset grid blocks are set up for the FANS code to carry out interpolations between donor and receiver points.

3.4. Numerical specifications

The details of the numerical specifications in the CFD calculations are provided in Table 3. It should be explained here that although slip-wall condition should be applied to the wall surfaces in an ideal scenario, they are instead specified as slippery, which is a compromise of computational resources. Although not presented in the current study, trial simulations with no-slip boundary conditions were also performed while the difference is negligible. For those mutually overlapped inner surfaces of overset grid blocks, the "interior boundary surfaces" are specified and interpolation of flow information will be conducted.

4. Quantification of discretization uncertainties in the CFD simulations

In this section, the previously introduced verification procedure is applied to quantify the numerical uncertainties in the CFD-predicted wake characteristics which stem from both the spatial and temporal discretization.

It should be emphasized that although not included in the present work, a systematic verification study assessing the errors and uncertainties in the calculated thrust and torque of this same turbine has already been conducted and reported in a previous publication by the authors (Ye et al., 2024b). Therefore, the current study focuses exclusively on the wake characteristics. Moreover, as demonstrated in our earlier work, the mesh resolution in the far-wake region—which is to be systematically refined here—has been shown to have negligible impact on the aerodynamic loads of the turbine (Ye et al., 2022a). Consequently, this study is confined to investigating the effect of mesh resolution on the wake characteristics, thereby performing a rigorous

Table 3
Boundary conditions in the FANS calculations.

Boundary name	Boundary type
Inlet	inflow
Outlet	zero gradient
Wind turbine wall surfaces	no-slip wall
Wind tunnel wall surfaces	slip wall
Inner surfaces of overset grid blocks	interior boundary surfaces

verification analysis specifically for the regions highlighted in yellow in Fig. 6.

In order to perform the verification study, a simulation matrix obtained by using 3 different computational grids and 3 different time increments is defined. The three computational grids as well as the 3 time increments adopted in this work are listed in the following two tables, respectively. The values of h_i and t_j are determined by the approach introduced in Eq. (11). In Table 4, the third and fourth columns are the number of grid points, in which the subscript "R" stands for "refined", and the subscript "T" represents "total". Recap that the grid blocks in the rotating domain, those grids covering the surface of the wind turbine, remain untouched.

Table 5 lists the details of the three computational time stepping sizes used to form the verification matrix, in which, the shortest and longest computational time steps are marked as T1 and T3, and the physical temporal duration within a computational time step is denoted by Δt_j .

As discussed in Section 2, the theoretical framework suggests that five distinct CFD predictions would suffice for solving Eq. (11). However, following recommendations by Eça and Hoekstra (2014) regarding least-squares fitting requirements, we incorporated additional data points to enhance solution robustness. To comprehensively investigate how numerical uncertainties in CFD-generated data influence neural network performance, we conducted simulations across a full matrix of nine grid-time configurations (h_i and t_j combinations). This systematic approach enables evaluation of spatial and temporal resolution impacts through three grid refinement levels (G1-G3) paired with three temporal increments (T1-T3). The resulting nine datasets - designated as G1T1, G1T2, G1T3, G2T1, G2T2, G2T3, G3T1, G3T2, and G3T3 - provide a controlled basis for assessing training data quality effects. These datasets subsequently serve as distinct training inputs for neural network development, allowing direct comparison of model performance across varying numerical resolution scenarios.

4.1. Iterative convergence

In the numerical simulation of the incompressible Navier-Stokes equations, apart from the inner loops which are used to solve the linear algebraic equations, outer iteration loops are needed to link, or correct, the pressure and velocity fields in an iterative manner to ensure the mass conservation of flow. In the simulations utilizing structured overset grid blocks, the authors find that the choice of the number of outer loops could be critical to the overall performance of the FANS code. On one hand, we wish the accuracy of the results to be as high as possible; on the other hand, larger number of outer loops will result into higher computational cost due to the fact that different overset blocks which stored in different processors will need to communicate with each other. Therefore, before we delve into the iterative residuals, a screen of the optimal outer loop number to be used in the following CFD simulations is performed. The outcome of the screening process is presented in Fig. 9(a). It could be seen that no further significant changes of the residuals after the outer iteration loop number reaches 20. Therefore, the number of outer iterations will be set as 20 in the current work.

With the use of 20 outer iterations, the iterative convergence is presented in Fig. 9(b). It can be seen from the figure that the residual of each individual equation remains at the order of 10^{-4} for most of the time. Although several spikes can be identified in the figure, those spikes are quickly smoothed and do not affect the overall accuracy of the simulation. According to Eq. (10), the iterative errors can be fairly

Table 4
Information of the computational grids.

Grids	h_i	N_R^i [million]	N_T^i [million]
G1	1.00	18.28	24.31
G2	1.20	10.58	16.77
G3	1.44	6.12	12.53

Table 5
List of the temporal resolution.

Time increments	t_j	Δt_j [s]	No. time steps per revolution
T1	1.0	0.00013	360
T2	2.0	0.00026	180
T3	3.0	0.00039	120

regarded as small quantities and thus will not be taken into consideration in the remainder of this paper.

4.2. Discretization uncertainty

With the premise of a careful discussion of iterative error, the discretization uncertainty in the CFD simulations can then be readily

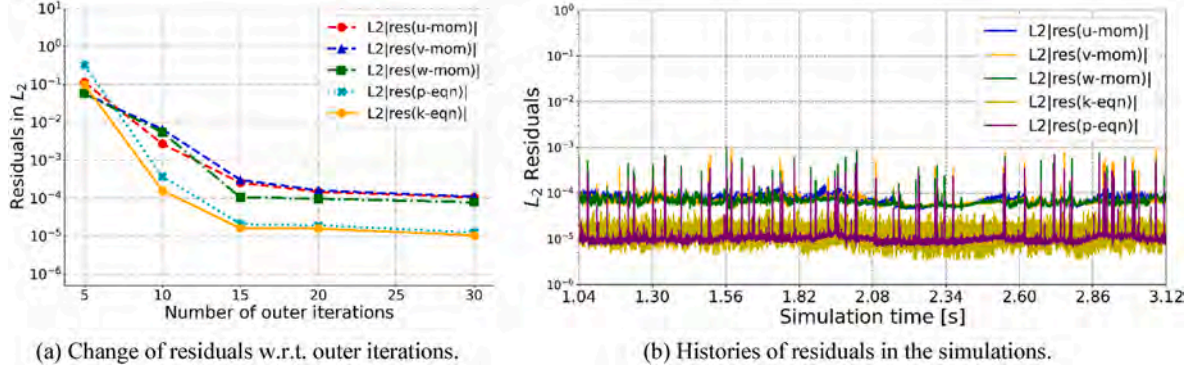


Fig. 9. Iterative convergence of the simulations.

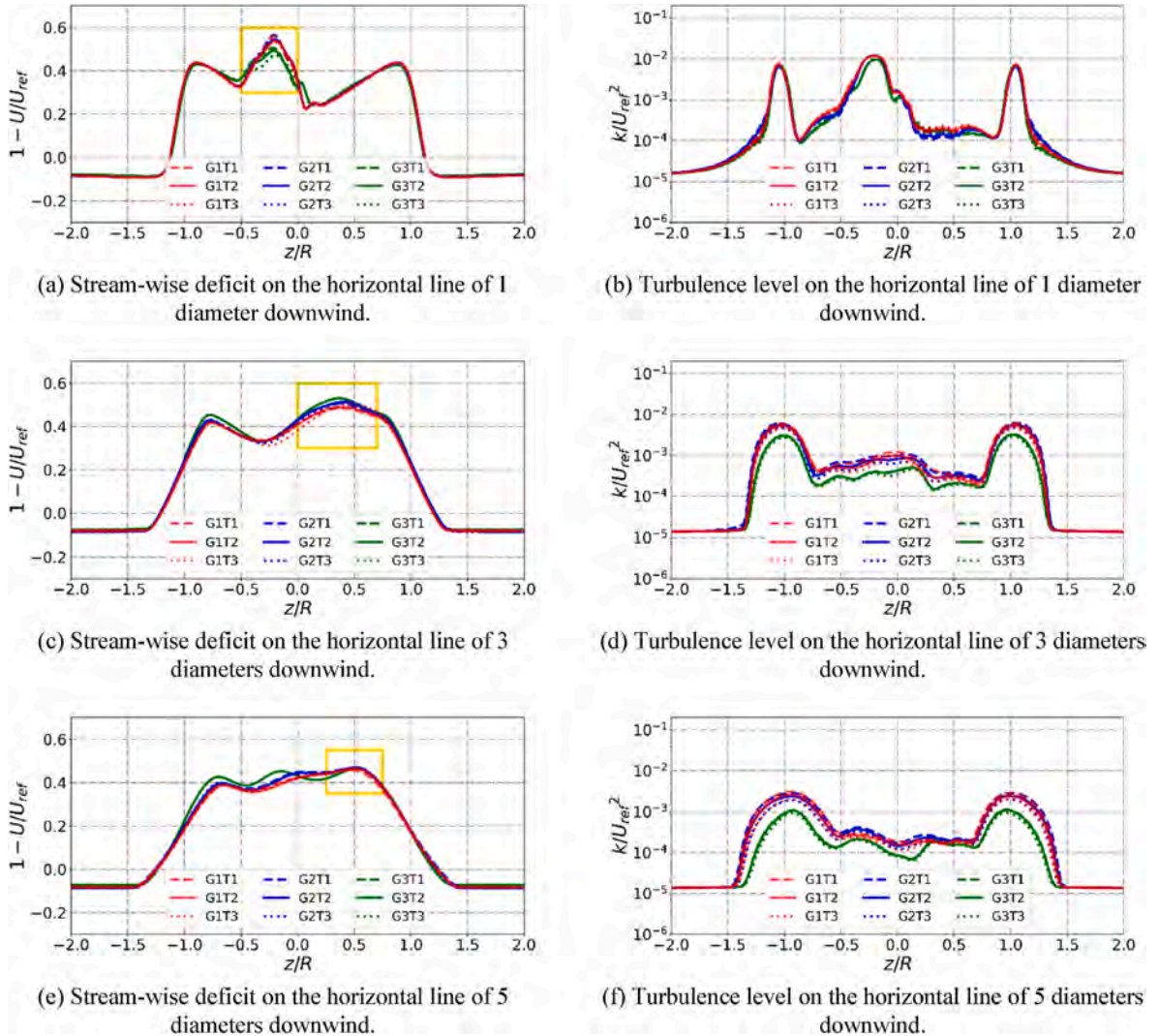


Fig. 10. Verification study of the CFD predictions of wake characteristics. The left column presents the deficit profiles while the right column presents the TKE profiles.

quantified by the adopted verification procedure. Fig. 10 presents the obtained wake profiles from all the 9 numerical configurations in the verification matrix. It should be pointed out here that the local maximums in the left-handed figures imply low magnitudes at given locations. For instance, in Fig. 10(a), two local maximums can be identified in the vicinity of $z/R = \pm 1.0$, and this implies that low-velocity regions are formed in this region (will be called “tip peaks” hereafter). However, an additional local maximum exists in the vicinity of $z/R = -0.25$, as marked by a cyan-color rectangle. This third local maximum forms because of the rotation and skew of the tower-generated wake, which is the key challenge in simulations of the wake of the selected wind tunnel test (Ye et al., 2023a). This local maximum will be called “asymmetry peak”.

The reason leading to the formation of this phenomenon, i.e. asymmetric wake profiles, has already thoroughly discussed by the authors in a previous work in which a traditional, or popular, numerical framework under the finite-volume (FV) method was adopted. The readers are encouraged to read the authors’ previous work (Ye et al., 2023a) for the thorough explanation. In the current work, the magnitudes of the asymmetry peaks obtained from the 9 numerical configurations are picked as the quantity of interest in the verification study, i.e. ϕ_{ij} in Eq. (11). As shown in the following figures, an orange-color rectangle is used to highlight the asymmetry peaks.

Subsequently, the asymmetric peaks are used as the metric in the verification study. Fig. 11 visualizes the results of the fitting procedure as introduced in Section 2. A curved plane is generated by means of least-squares fitting for the numerical predictions at each of the three downstream locations, i.e. 1 D, 3 D, and 5 D, and thus the “exact” solution of the numerical prediction, ϕ_0 in Eq. (11), as if infinite resolutions of grid and time increment were used can then be determined by setting $h_i = t_j = 0$. Afterward, the discretization error or uncertainty of any of the 9 CFD predictions can be estimated with respect to ϕ_0 .

Table 6 summarizes the CFD-predicted wake profiles at 1 D downwind by using different pairs of h_i and t_j , i.e. ϕ_{ij} , the absolute errors between ϕ_{ij} and ϕ_0 , i.e. ϵ_ϕ (in percentage), and the estimated uncertainty of ϕ_{ij} , i.e. U_ϕ (in percentage). And Tables 7 and 8 are the summaries for 3 D and 5 D downwind locations, respectively. For example, the CFD-predicted peak wake deficit obtained by using G1T2 at 1 D downstream is 0.529, and its absolute error is 10.6 %, and the corresponding numerical uncertainty is 34.5 %.

The numerical uncertainty analysis results presented in Tables 6–8 are visually summarized in the bar charts of Fig. 12(a)–(c). Each bar represents CFD-predicted velocity deficit values, with black error bars indicating the corresponding numerical uncertainty ranges. The blue dashed line denotes the theoretical solution (ϕ_0) derived through least-squares fitting of all the 9 CFD datasets. These figures demonstrate clearly that maximum numerical uncertainties (U_ϕ) occur at 1D

Table 61D downstream, $\phi_{ii}/\epsilon_\phi/U_\phi \cdot \phi_0 = 0.592$

	T1	T2	T3
G1	0.549/7.3/25.1	0.548/7.4/34.1	0.502/15.2/45.1
G2	0.529/10.6/34.5	0.527/11.0/43.0	0.493/16.7/53.2
G3	0.502/15.2/48.6	0.494/16.7/55.4	0.473/20.1/67.0

Table 73D downstream, $\phi_{ii}/\epsilon_\phi/U_\phi \cdot \phi_0 = 0.441$

	T1	T2	T3
G1	0.473/7.3/19.9	0.476/7.9/22.4	0.483/9.5/27.0
G2	0.484/9.8/28.0	0.493/11.8/31.8	0.498/12.9/35.5
G3	0.503/14.1/39.6	0.507/15.0/42.3	0.514/16.6/46.9

Table 85D downstream, $\phi_{ii}/\epsilon_\phi/U_\phi \cdot \phi_0 = 0.479$

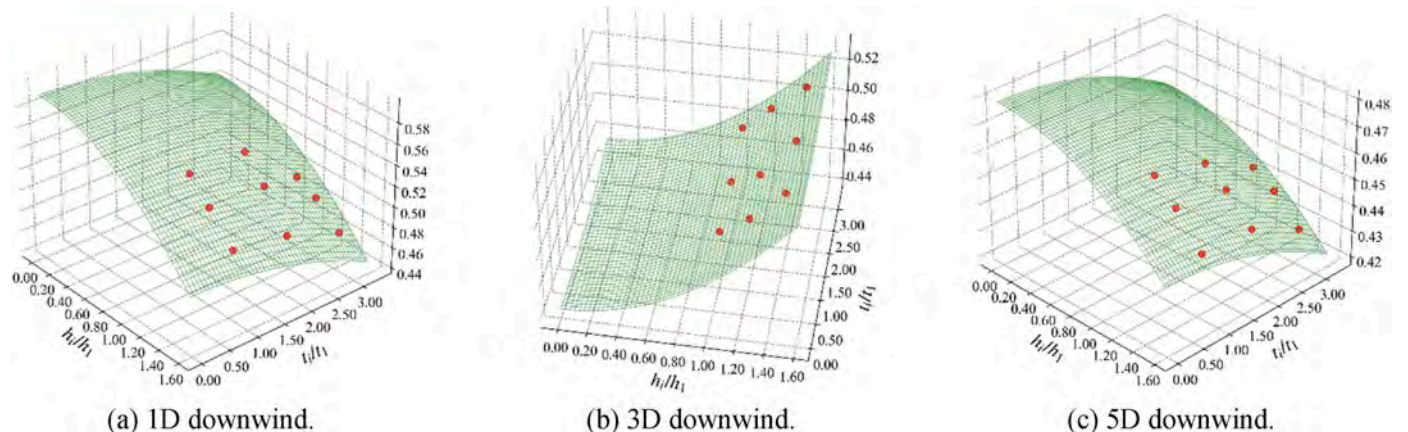
	T1	T2	T3
G1	0.462/3.5/10.9	0.457/4.6/14.8	0.446/6.9/21.3
G2	0.454/5.2/15.5	0.451/5.9/19.0	0.441/7.9/25.4
G3	0.442/7.7/22.0	0.441/7.9/24.8	0.431/10.0/31.3

downstream, gradually decreasing to 3D and 5D positions. This spatial attenuation pattern aligns with previous findings from the authors’ previous work (Ye et al., 2023a), where similar uncertainty reduction trends were observed using a finite-volume method with sliding-interface techniques. A detailed discussion of this phenomenon is beyond the scope of this paper, and interested readers are referred to the cited work for comprehensive analysis of the underlying mechanisms driving this uncertainty reduction as downstream distance increases.

4.3. Validation of CFD results

The deficit profiles obtained by the numerical configuration of G1T1 in the verification matrix are plotted in the same figure with the experimental data (black squares) for comparison, as shown in the following figures. Current results are indicated by red square. In addition to the comparison of modeling approaches presented here, results of selected numerical frameworks as described in (Krogstad and Eriksen, 2013), (Ye et al., 2023a), (Ji et al., 2022) are also included and detailed in Table 9.

It can be observed from Fig. 13 that the deficit profiles obtained in this work align closely with the experimental measurement. However, for the TKE profiles large deviations in magnitudes between the present CFD predictions and the experimental measurements can be identified.

**Fig. 11.** Illustration of the least-squares fitting in the verification procedure.

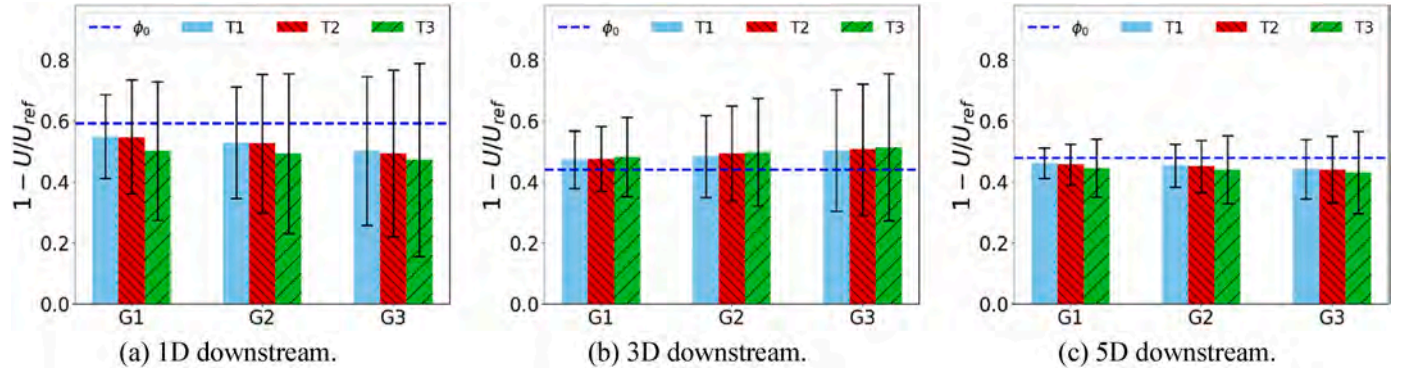


Fig. 12. CFD results and their corresponding uncertainties at different downstream distances.

Table 9

Selected representative numerical results to be compared with.

Reference Name	Rotor	Tower	Gov. PDEs	Discretization	Code
FRG-RANS w/o tower	FRG	Not considered	RANS	FV	STAR CCM+
AL-LES	AL	Not considered	LES	FV	EllipSys3D
AL-IB-LES	AL	Considered (IB)	LES	FV	OpenFOAM
ReFresco	FRG	Considered	RANS	FV	ReFresco

FRG: fully-resolved geometry; AL: actuator line; IB: immersed body.

- FRG-RANS w/o tower: RANS simulations with fully-resolved wind turbine blade geometry but without the tower.
- AL-LES: LES simulations with the rotor modeled by the actuator line method, but the tower was not considered.
- AL-IB-LES: LES simulations with the rotor modeled by the actuator line method, and the tower was modeled by the immersed body method.
- ReFresco: RANS simulations with fully-resolved wind turbine blade geometry and the tower structure.

This phenomenon was also observed by (Ye et al., 2023a), i.e. the results shown in blue lines and marked as ReFresco. As discussed by (Ye et al., 2023a), this discrepancy could be owing to the limitations of eddy-viscosity based models, i.e. $k - \epsilon$ turbulence model in the present study, in which an isotropic assumption of P_k is adopted. Nevertheless, it should also be recognized that even though the TKE levels are under-predicted, the overall characteristics of the profiles of the current CFD prediction agree with the experimental measurements reasonably well.

5. Training of NN models for wind-turbine wake predictions

The 9 CFD datasets with varying numerical uncertainties (G1T1, G1T2, G1T3, G2T1, G2T2, G2T3, G3T1, G3T2, G3T3) are then used to train the weakly-constrained PINN to evaluate the impact of training data quality. Notably, while the network structure remained identical across all cases, each model was trained exclusively on one dataset and subsequently named after its respective dataset (e.g., G1T1 denotes the model trained on Grid 1 with T1). This approach isolates the influence of numerical uncertainties in CFD data on model performance.

5.1. Definition of NN mapping

The weakly-constrained PINN establishes the following mapping between spatiotemporal coordinates and velocity components:

$$(u, v, w) = \text{PINN}(x, y, z, t) \quad (15)$$

Here, u , v , and w represent the velocity components at spatial coordinates (x, y, z) and time t . This formulation enables the network to predict instantaneous wake velocity fields across the entire domain using sparse input data.

5.2. Data pre-processing

Prior to training, the dataset was preprocessed to enhance numerical stability and training efficiency. Data points were randomly shuffled to eliminate positional bias, and min-max normalization was applied to scale all input features to the range $[0, 1]$:

$$x_{\text{normalized}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (16)$$

This normalization ensures consistent feature scaling across the spatiotemporal domain, mitigating gradient-related issues during optimization.

5.3. Training specifications

The network architecture comprised 11 hidden layers with 100 neurons per layer, utilizing the hyperbolic tangent ($\tanh$) activation function:

$$a_{\tanh} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (17)$$

Training was implemented in PyTorch using the Adamax optimizer with a batch size of 64 and a learning rate of 0.0001. The loss function is defined by the mean squared error (MSE). In addition, a critical hyperparameter is the weighting factor ($factor$) balancing data-driven ($Loss_{\text{label}}$) and physics-based ($Loss_{\text{pde}}$) losses. While prior studies propose adaptive $factor$ tuning (Han et al., 2022), (Niu et al., 2025), we adopted a rather straight-forward and direct approach in the current study: the value of $factor$ is manually adjusted during training (e.g., be checked every 100 epochs) to ensure simultaneous reduction of both loss components. If either loss exhibited divergence, $factor$ was recalibrated to restore stable optimization. This strategy avoids overfitting to either data or physics constraints, maintaining balanced convergence.

6. Results and discussions

6.1. Comparison of model training efficiency

Fig. 14 illustrates the evolution of loss functions during 5000 training epochs for neural networks (NNs) trained on datasets with varying numerical uncertainties. The label loss (Fig. 14(a)) exhibits similar convergence rates across all datasets, indicating comparable data-fitting capabilities regardless of spatial or temporal uncertainties of the CFD data. However, the physics-informed (PDE) loss (Fig. 14(b)) reveals distinct trends: models trained on datasets with high numerical uncertainties, in particular low spatial resolutions (G3T1, G3T2, G3T3), show significantly slower PDE loss reduction compared to those using CFD data with lower numerical uncertainty (high spatial resolution, i.e. G1 and G2). This suggests that high spatial uncertainty, i.e. coarse spatial resolution, in training data significantly undermines the NN's

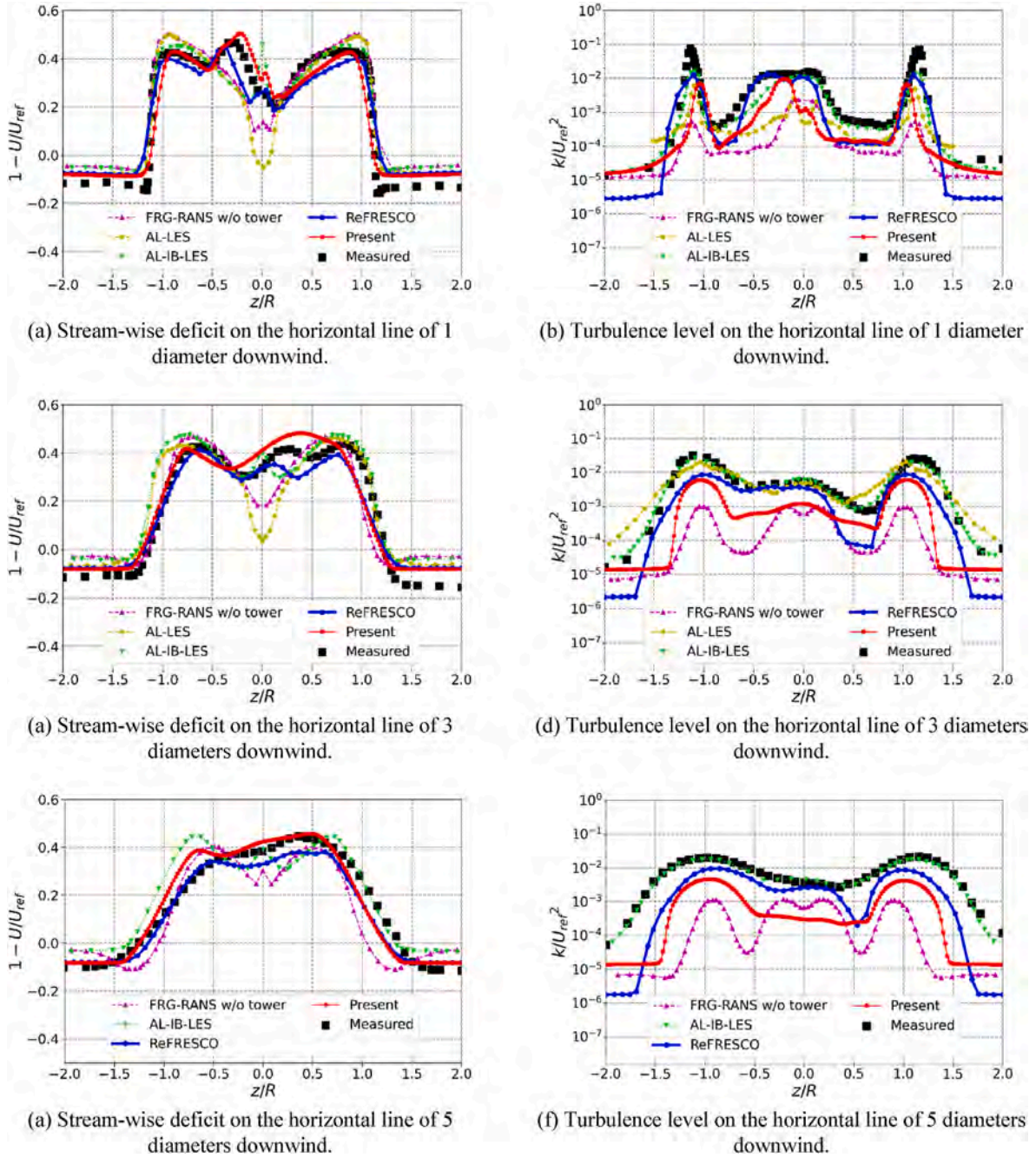


Fig. 13. Validation of CFD results.

ability to learn underlying physical constraints effectively.

6.2. Model accuracy on sparse wake lines

The predictive performances of the 9 NN models are evaluated on three sparse measurement lines (1D, 3D, and 5D downstream) using training and testing datasets. Fig. 15 compares NN-predicted velocity profiles (lines) against CFD ground truth (black circles). Both training (left column) and testing (right column) predictions demonstrate strong alignment with reference data across all downstream locations.

The black circles in Fig. 16 represent the ground truth (CFD results). The prediction accuracy was quantified using the coefficient of determination (R^2) which is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - y_i^{pred})^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (18)$$

Table 10 summarizes the R^2 scores for wake profile reconstruction at 1D, 3D, and 5D downstream for the 9 models. Fig. 16(a)–(c) visualize these results.

According to Table 10 and Fig. 16, following observations can be summarized.

- Regardless of the temporal resolution, datasets from G1 and G2 grids yielded NN models with R^2 values close to 1, indicating excellent prediction accuracy across all downstream distances.
- Limitation of low spatial resolution: Datasets from G3 grids resulted in significantly lower R^2 values (e.g., below 0.9 at 1D), highlighting

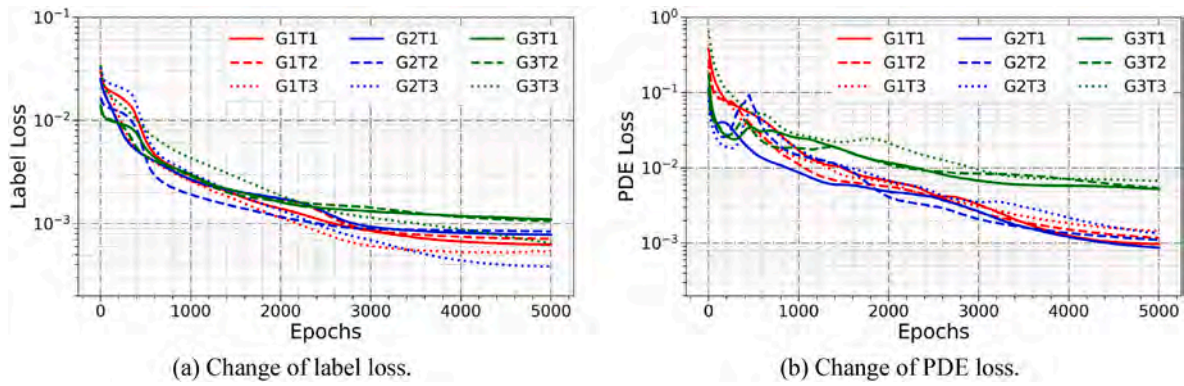


Fig. 14. Change of losses in the training process.

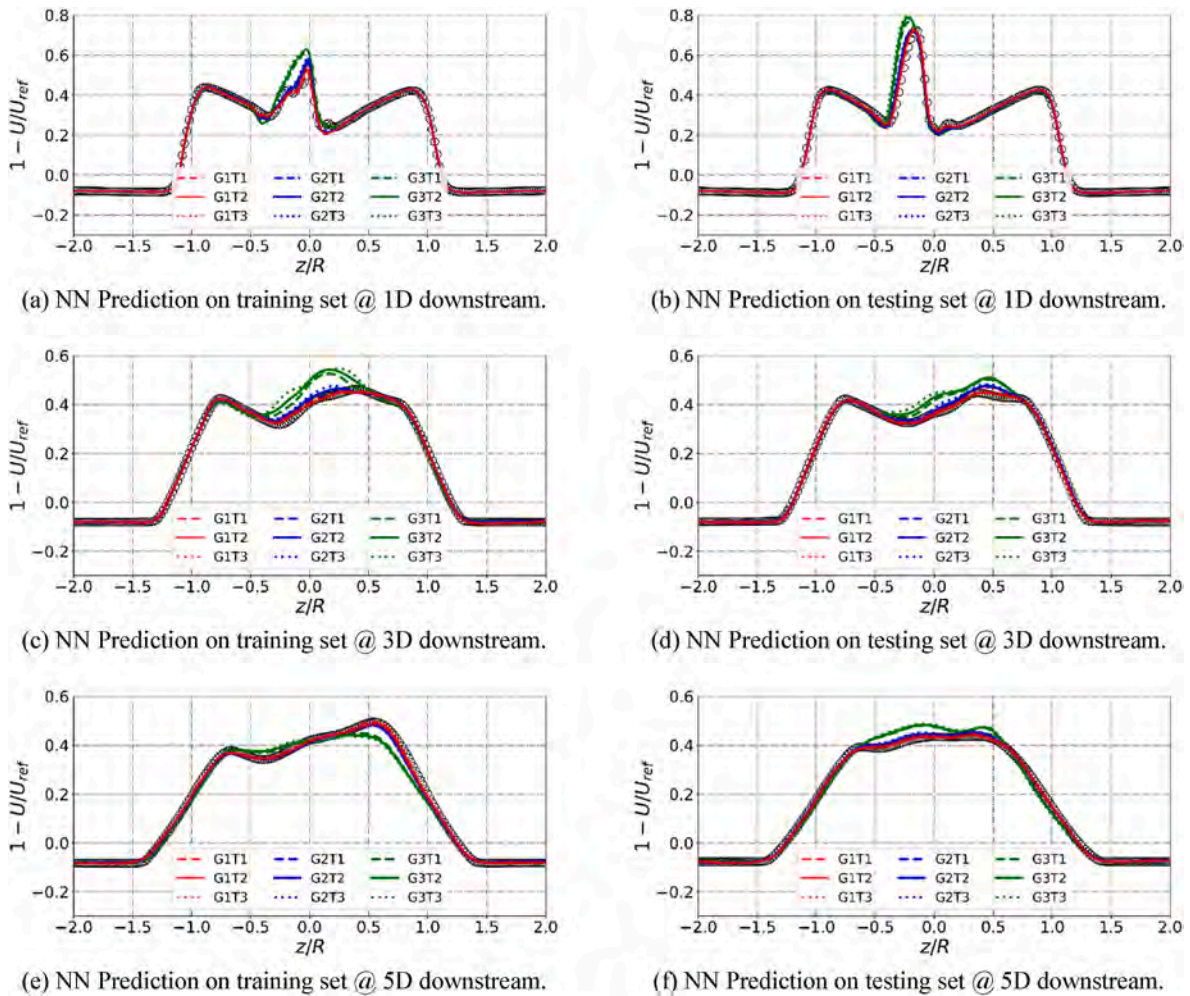


Fig. 15. Wake profiles predicted by different NN models. The left-column figures represent the training-set predictions, and the right-column figures represent the testing-set predictions.

the critical impact of spatial resolution on the performances of trained NN models.

- Downstream distance effect: Prediction accuracy increased with downstream distance for all models, with far-field (5D) predictions outperforming near-field (1D) results. This aligns with reduced wake asymmetry in the far field.
- Grid resolution vs. time increment: The difference in accuracy between G1 and G2 models diminished at greater downstream

distances, while time increment had a minimal impact on wake profile reconstruction accuracy.

6.3. Effect of numerical uncertainty on the full-wake construction

After evaluating performance on sparse measurement lines, the 9 NN models were used for full-wake prediction to further assess the impact of numerical uncertainties. It is worth mentioning that in the current work, we will limit ourselves to the wake prediction at hub height due to the

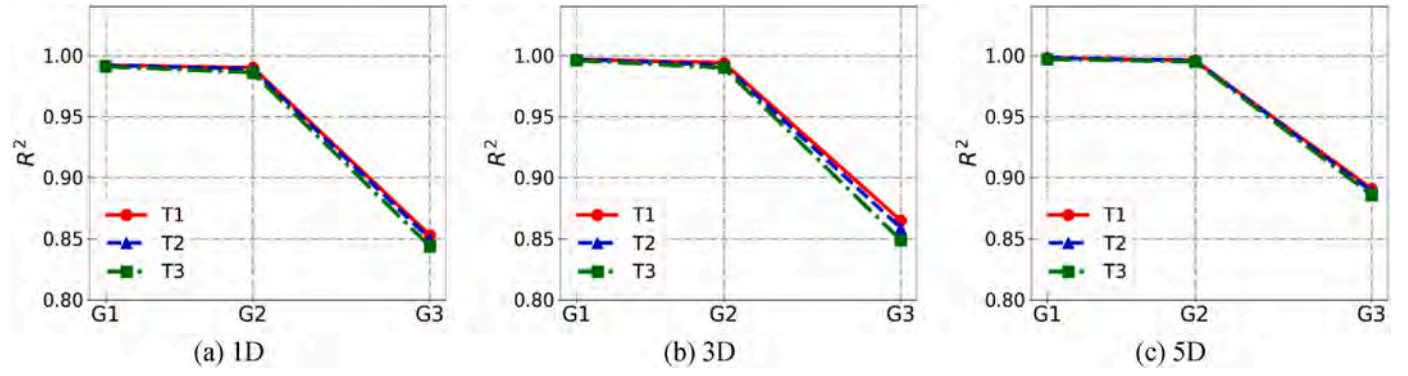


Fig. 16. R^2 scores of different NN models in the wake profile predictions at different downstream distances.

Table 10

R^2 scores of different NN models in the wake profile predictions.

Each box has three values which is corresponding to the R^2 score of the prediction at 1D, 3D, and 5D downstream.

	T1	T2	T3
G1	0.992/0.997/0.998	0.991/0.997/0.998	0.991/0.996/0.997
G2	0.990/0.994/0.996	0.989/0.992/0.996	0.986/0.990/0.995
G3	0.853/0.865/0.891	0.850/0.859/0.889	0.844/0.849/0.886

following reasons: (1) wind farm wake assessments typically require 2D velocity distributions at hub height; (2) 3D wake prediction is more complex, and the study prioritizes evaluating CFD data quality on NN performance in a manageable 2D context. The scope of the current paper is to discuss the influence of the numerical uncertainties in the CFD-generated training data on the final performance of the NN models. Therefore, prediction of the hub-height velocity of the wind turbine

wake should be sufficient to achieve this goal. Although we do expect that the in three-dimensional wind-turbine-wake prediction tasks, this influence should be much more pronounced.

Fig. 17 shows the predicted full wake fields by using the 9 trained NN models, with black-dashed lines indicating the location of data used to train the models. In general, the NN models trained on G1 and G2 datasets (Fig. 17(a)–(f)) accurately captured the overall wake characteristics and asymmetric features, while those trained on G3 datasets (Fig. 17(g)) produced significant discrepancies, failing to reproduce near-field asymmetry.

Further quantitative evaluation via R^2 scores (Table 11 and Fig. 18) revealed:

- The NN models which were trained by using G1/G2 grids, including G1T1, G1T2, G1T3, G2T1, G2T2 and G2T3, achieve high full-field prediction accuracy ($R^2 > 0.88$) regardless of temporal resolution.

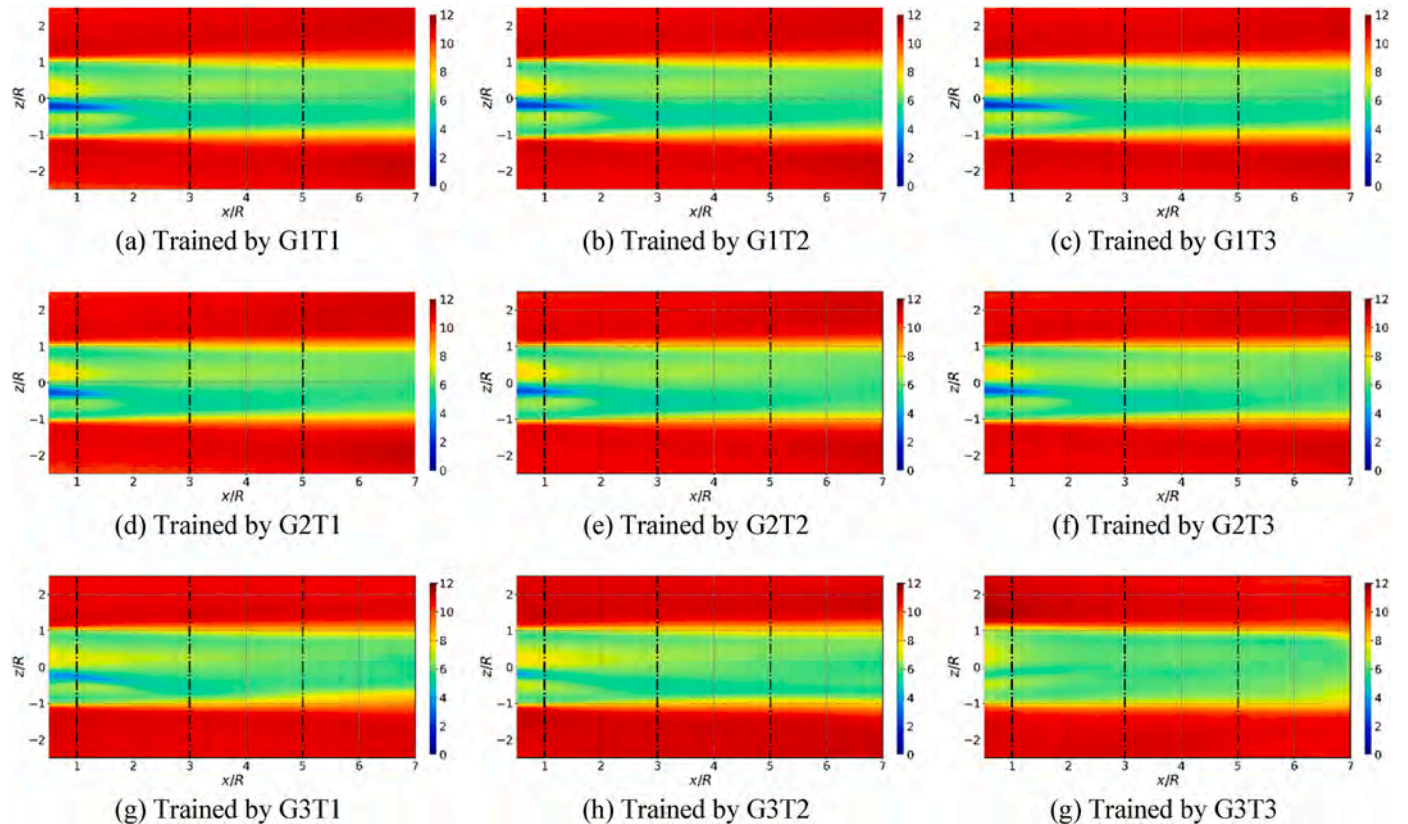


Fig. 17. Full-wake predictions by different NN models.

Table 11
R² scores of the full-wake predictions.

	T1	T2	T3
G1	0.91232	0.90763	0.90465
G2	0.89348	0.88103	0.87832
G3	0.78203	0.73394	0.69939

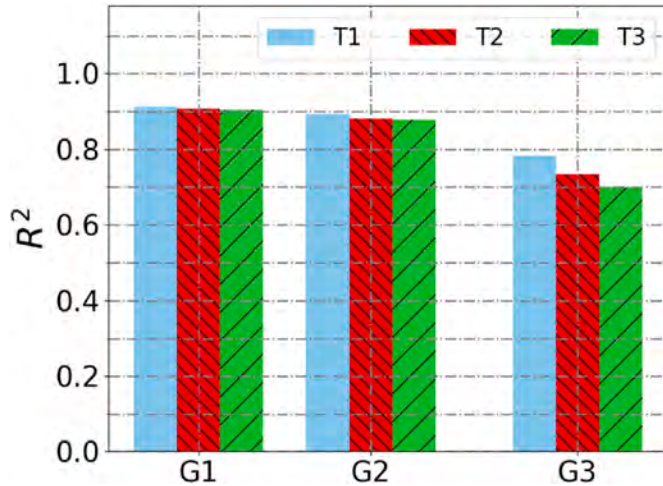


Fig. 18. Comparison of R² scores of different NN models in the full-wake prediction task.

- The NN models which were trained by using G3 grid, including G3T1, G3T2, G3T3, show severe performance degradation ($R^2 < 0.78$), with temporal resolution effects becoming more pronounced compared to sparse-line prediction tasks.
- Practical recommendation: G2T3 configuration balances accuracy ($R^2 = 0.878$) and computational efficiency, achieving 95 % of G1T1 performance at reduced cost.
- According to the above observations, it should be legitimate to recommend to use moderate-resolution grids (e.g., G1T3 or G2T3) in the data-generation phase by CFD computations. This approach can provide a viable balance between accuracy and computational cost, maintaining R^2 above 0.87 while reducing grid points compared to high-resolution configurations like G1T1.

Those above results emphasize that spatial resolution in CFD training data critically determines NN model quality, particularly for wake reconstruction tasks by only using training data collected at sparse lines. Temporal resolution plays a secondary role, suggesting prioritization of spatial refinement during dataset generation.

7. Conclusions

This study investigates how numerical uncertainties in CFD-generated training data - particularly spatial and temporal discretization errors - affect the final performances of trained neural networks (NNs) in predicting wind turbine wake fields from sparse sensor measurements. Given that training data for NN models of wake prediction is inherently limited to sparse points (e.g., lidar lines), numerical uncertainties in CFD datasets can significantly degrade the accuracy of trained models.

First, a numerical framework combining the finite-analytic (FA) method for discretizing unsteady RANS equations with overset-grid capabilities was employed to simulate wind turbine wakes, generating training data for NN models. Then, a systematic verification procedure quantified spatial and temporal discretization errors using a matrix of nine CFD configurations (varying grid resolutions and time increments).

Meanwhile, a "weakly-constrained PINN" model was developed to train on sparse wake data. Nine NN models were trained using CFD datasets with different uncertainty levels, and their performance was evaluated on two tasks: (1) predicting wake profiles at sparse measurement lines (1D, 3D, 5D downstream), and (2) predicting velocity fields at wind turbine hub height. Key conclusions include.

1. Velocity profile predictions at sparse lines

- Temporal resolution impact: For the same grid resolution, CFD training data with different time increments yielded NN models with comparable performance, indicating minimal sensitivity to temporal discretization errors.
- Downstream distance effect: prediction errors decreased by 8 %–12 % from 1D to 5D across all configurations, as wake asymmetry diminishes with distance.
- Spatial resolution criticality: datasets from the coarsest grid (G3T1, G3T2, G3T3) resulted in significantly poorer prediction accuracy (e.g., low R^2 values at 1D), highlighting the dominant role of spatial discretization accuracy in CFD-generated training data quality.

2. Full-wake prediction

- High-resolution data advantage: High-resolution grids (G1/G2) enabled higher prediction accuracies, capturing asymmetric wake features (e.g., tower-induced deficits) with an overall R^2 score greater than 0.85. Low-resolution grids (G3) failed to reproduce these details.
- Cost-performance balance: sing moderate-resolution grids (G2 configurations) offered a viable balance between computational cost and reconstruction accuracy, reducing resource demands while maintaining acceptable performance.

Based on these findings, we propose the following guidelines for CFD-generated training data in wake modeling.

- 1) Systematic Verification: Generate training datasets using a simulation matrix with varied grid/time-increment configurations to quantify numerical uncertainties;
- 2) NN Sensitivity Testing: Train identical NN architectures on datasets of differing accuracy to assess the impact of numerical uncertainty of the training data on the final performance of the trained NN models;
- 3) Determination of Spatial and Temporal Resolutions in CFD simulations: Use the accuracy-resolution curve from step 2 to determine the minimum CFD resolution required for acceptable trained NN performance, balancing data generation consumption and model accuracy.

Our study underscores the importance of integrating uncertainty quantification into data-driven wake modeling, providing a framework for efficient and reliable neural network training in wind energy applications. Future work should extend this framework to 3D wake modeling.

CRediT authorship contribution statement

Maokun Ye: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Yao Zhong:** Visualization, Validation, Investigation, Formal analysis, Data curation. **Hamn-Ching Chen:** Writing – review & editing, Validation, Investigation, Formal analysis, Conceptualization. **Decheng Wan:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (52131102), to which the authors are most grateful.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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